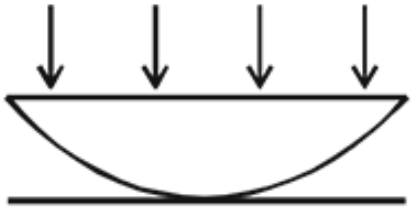
Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

A thin slice is cut out of a glass cylinder along a plane parallel to its axis. The slice is placed on a flat glass plate as shown in Figure. The observed interference fringes from this combination shall be



(A) straight

(B) circular

(C) equally spaced

(D) having fringe spacing which increases as we go outwards

Q.2

Two monochromatic light beams of intensity 16 and 9 units are interfering. The ratio of intensities of bright and dark parts of the resultant pattern is:

(A) $\frac{16}{9}$

(B) $\frac{4}{3}$

(C) $\frac{7}{1}$

(D) $\frac{49}{1}$

Q.3

A thin sheet of mica $\left(12 \times 10^{-7} \mathrm{~m} \text{ thick }\right)$ is placed in the path of one of the interfering beams in a biprism arrangement. It is found that the central bright band shifts a distance equal to the width of a bright fringe. The refractive index of mica $\left(\text{ Given } \lambda=6 \times 10^{-7} \mathrm{~m}\right)$ is

(A) 1.0

(B) 1.5

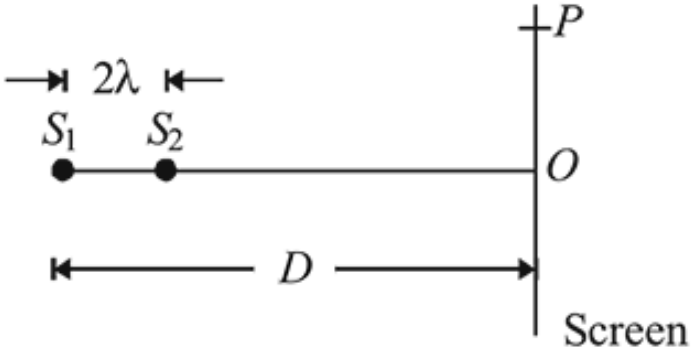
(C) 1.75

(D) 1.25

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Q.4

Two sources S_1 and S_2 emitting coherent light waves of wavelength $\bar{\lambda}$ in the same phase are situated as shown. The distance OP, so that the light intensity detected at P is equal to that at O is



(A) $D\sqrt{2}$

(B) $D/2$

(C) $D\sqrt{3}$

(D) $D/\sqrt{3}$

Q.5

A plastic sheet (refractive index $= 1.6$) covers one slit of a double slit arrangement meant for the Young's experiment. When the double slit is illuminated by monochromatic light (wavelength in air $=$ $6600\mathring{A}$), the centre of the screen appears dark rather than bright. The minimum thickness of the plastic sheet to be used for this to happen is:

(A) $3300\mathring{A}$

(B) $6600\mathring{A}$

(C) $2062\mathring{A}$

(D) $5500\mathring{A}$

Q.6

A plane wave of monochromatic light falls normally on a uniform thin layer of oil which covers a glass plate. The wavelength of source can be varies continuosly. Complete destructive interference is observed for $\lambda = 5000\mathring{A}$ and $\lambda = 1000\mathring{A}$ and for no other wavelength in between. If μ of oil is 1.3 and that of glass is 1.5, the thickness of the film will be

(A) 6.738×10^{-5} cm

(B) 5.7×10^{-5} cm

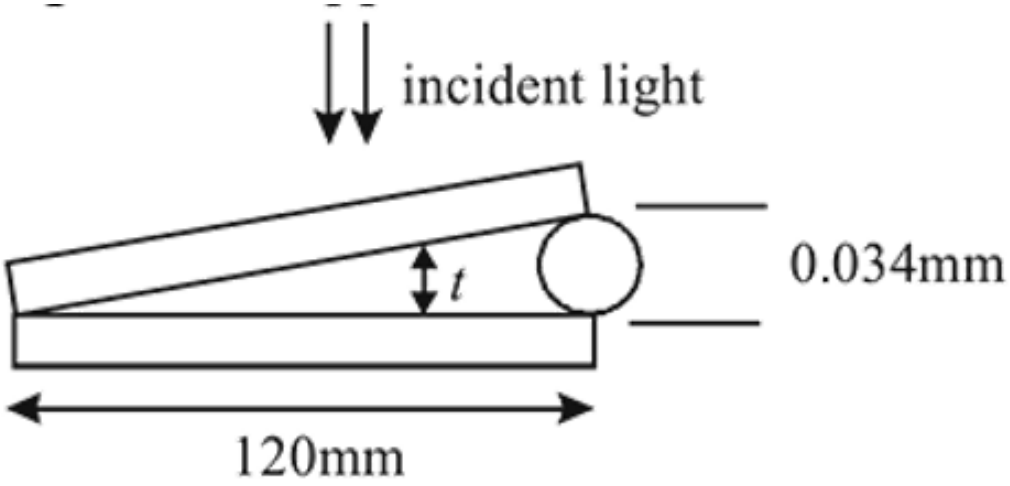
(C) 4×10^{-5} cm

(D) 2.8×10^{-5} cm

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Q.7

A broad source of light $(I = 680\ \text{nm})$ illuminates normally two glass plates $120\ \text{mm}$ long that touch at one end and are separated by a wire $0.034\ \text{mm}$ in diameter at the other end. The total number of bright fringes that appear over the $120\ \text{mm}$ distance is



(A) 50

(B) 100

(C) 200

(D) 400

Q.8

Two light waves having the same wavelength λ in vacuum are in phase initially. Then the first ray travels a path of length L_1 through a medium of refractive index μ_1. The second ray travels a path of length L_2 through a medium of refractive index μ_2. The two waves are then combined to observe interference effects. The phase difference between the two, when they interfere, is

(A) $\frac{2\pi}{\lambda}\left(\mathrm{L}_1 - \mathrm{L}_2\right)$

(B) $\frac{2\pi}{\lambda}\left(\mu_1\,\mathrm{L}_1 - \mu_2\,\mathrm{L}_2\right)$

(C) $\frac{2\pi}{\lambda}\left(\mu_2\,\mathrm{L}_1 - \mu_1\,\mathrm{L}_2\right)$

(D) $\frac{2\pi}{\lambda}\left[\frac{\mathrm{L}_1}{\mu_1} - \frac{\mathrm{L}_2}{\mu_2}\right]$

Q.9

A light of wavelength $6000\mathring{A}$ shines on two narrow slits separated by a distance $1.0\ \text{mm}$ and illuminates a screen at a distance $1.5\ \text{m}$ away. When one slit is covered by a thin glass plate of refractive index 1.8 and other slit by a thin glass plate of refractive index μ, the central maxima shifts by 0.1rad. Both plates have the same thickness of $0.5\ \text{mm}$. The value of refractive index μ of the glass is

(A) 1.4

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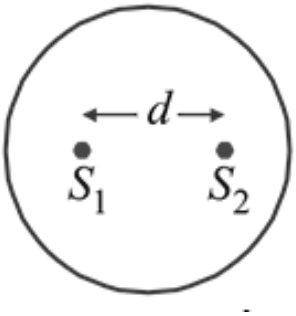
(B) 1.5

(C) 1.6

(D) None of these

Q.10

Two coherent sources separated by distance a are radiating in phase having wavelength λ. A detector moves in a big circle around the two sources in the plane of the two sources. The angular position of $n = 4$ interference maxima is given as



(A) $\sin^{-1}\frac{n\lambda}{d}$

(B) $\cos^{-1}\frac{4\lambda}{d}$

(C) $\tan^{-1}\frac{d}{4\lambda}$

(D) $\cos^{-1}\frac{\lambda}{4d}$

Q.11

In an experiment, sodium light $(\lambda = 5890\ \text{A}^\circ)$ is employed and interference fringes are obtained in which 20 fringes equally spaced occupy $2.30\ \text{cm}$ on the screen. When sodium light is replaced by blue light, the setup remaining the same otherwise, 30 fringes occupy $2.80\ \text{cm}$. The wavelength of blue light is

(A) $4780\mathring{A}$

(B) $5760\mathring{A}$

(C) $9720\mathring{A}$

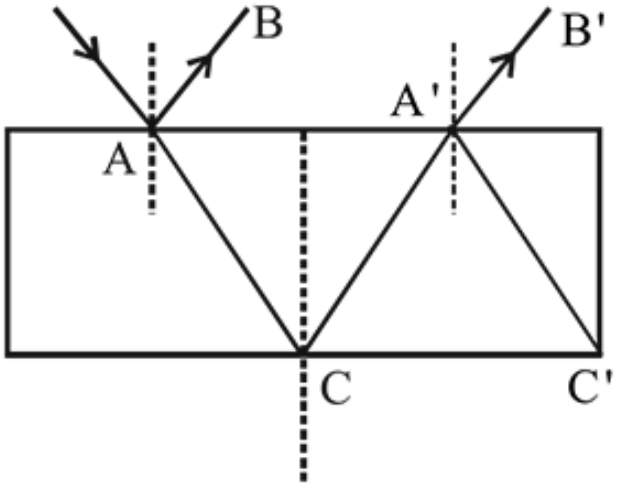
(D) $6390\mathring{A}$

Q.12

A ray of light of intensity I is incident on a parallel glass slab at point A as shown in diagram. It undergoes partial reflection and refraction. At each reflection, 25% of incident energy is reflected. The

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rays AB and $A'B'$ undergo interference. The ratio of I_{max} and I_{min} is :



(A) $49 : 1$

(B) $7 : 1$

(C) $4 : 1$

(D) $8 : 1$

Q.13

A wedged shaped air film having an angle of 40 second is illuminated by a monochromatic light and the fringes are observed vertically down through a microscope. The fringe separation between two consecutive bright fringes is $0.12\ \text{cm}$. The wavelength of light is:

(A) $5545\mathring{A}$

(B) $6025\mathring{A}$

(C) $4925\mathring{A}$

(D) $4655\mathring{A}$

Q.14

Two identical coherent sources are placed on a diameter of a circle of radius R at separation x $(\ll < R)$ symmetrical about the center of the circle. The sources emit identical wavelength λ each. The number of points on the circle of maximum intersity is $(x = 5\lambda)$

(A) 20

(B) 22

(C) 24

(D) 26

Q.15

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To produce a minimum reflection of wavelengths near the middle of visible spectrum $(550\ \text{nm})$, how thick should a coating of $MgF_2(\mu = 1.38)$ coated on a glass surface?

(A) 10^{-7}

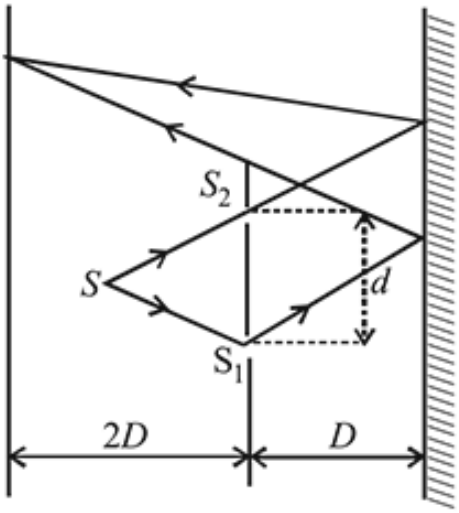
(B) 10^{-10}

(C) $10^{-9}\ \text{m}$

(D) $10^{-8}\ \text{m}$

Q.16

A double slit, $S_1 - S_2$ is illuminated by a light source S emitting light of wavelength λ. The slits are separated by a distance d. A plane mirror is placed at a distance D in front of the slits and a screen is placed at a distance $2D$ behind the slits. The screen receives light reflected only by the plane mirror. The fringe-width of the interference
pattern on the screen is



(A) $\frac{D\lambda}{d}$

(B) $\frac{2D\lambda}{d}$

(C) $\frac{3D\lambda}{d}$

(D) $\frac{4D\lambda}{d}$

Q.17

Two rectangular glass plates are in contact at one edge while the other edges are separated by a space of some suitable thickness so as to form a low angle wedge. The spacer is placed parallel to the line of contact and is at a distance of $10\ \text{cm}$ from it. When viewed normally in light of wavelength $5500\mathring{A}$, a series of evenly spaced dark bands $0.5\ \text{mm}$ apart are seen. The thickness
of the spacer is :

(A) $0.0425\ \text{cm}$

(B) $0.0036\ \text{cm}$

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(C) 0.0055 cm

(D) 0.0254 cm

Q.18

Assuming human pupil to have a radius of 0.25 cm and a comfortable viewing distance of 25 cm, the minimum separation between two objects that human eye can resolve at 500 nm wavelength is

(A) 100μm

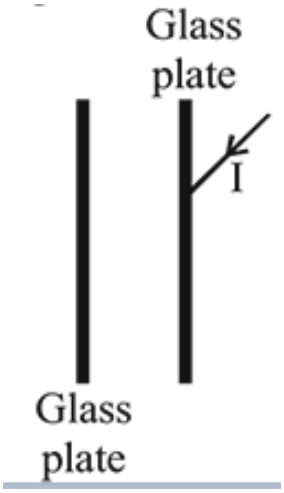
(B) 300μm

(C) 1μm

(D) 30μm

Q.19

A narrow monochromatic beam of light of intensity I is incident on a glass plate as shown in figure. Another identical glass plate is kept close to the first one and parallel to it. Each glass plate reflects 25 per cent of the light incident on it and transmits the remaining. Find the ratio of the minimum and the maximum intensities in the interference pattern formed by the two beams obtained after one reflection at each plate.



(A) $1:30$

(B) $1:49$

(C) $1:51$

(D) $1:73$

Q.20

Which one of the following phenomenon is not explained by Huygen's construction of wavefront?

(A) Refraction

(B) Reflection

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(C) Diffraction

(D) Origin of spectra

Answers & Solutions

Q.1 Answer:

straight

Solution:

Locus of equal path difference are lines running parallel to axis of the cylinder. Hence straight fringes will be observed.

Q.2 Answer:

$\frac{49}{1}$

Solution:

Intensity $\propto$ **(amplitude)** ^{2} $\frac{I_1}{I_2} = \frac{16}{9} = \frac{a_1^2}{a_2^2} \Rightarrow \mathrm{a_1} = 4; \mathrm{a_2} = 3$

Therefore the ratio of intensities of bright and dark parts $\frac{I_{\text{Bright}}}{I_{\text{Dark}}} = \frac{(a_1+a_2)^2}{(a_1-a_2)^2} = \frac{(4+3)^2}{(4-3)^2} = \frac{49}{1}$

Q.3 Answer:

1.5

Solution:

Given: $\mathrm{t} = 12 \times 10^{-7}\ \mathrm{m}, \lambda = 6 \times 10^{-7}\ \mathrm{m}, \mathrm{n} = 1, \mu = ?$

The shifts,

$$(\mu - 1)\mathrm{t} = \mathrm{n}\lambda$$
$$\mu = \frac{\mathrm{n}\lambda}{\mathrm{t}} + 1 = \frac{1 \times 6 \times 10^{-7}}{12 \times 10^{-7}} + 1 = 1.5$$

The refractive index of mica $= 1.5$

Q.4 Answer:

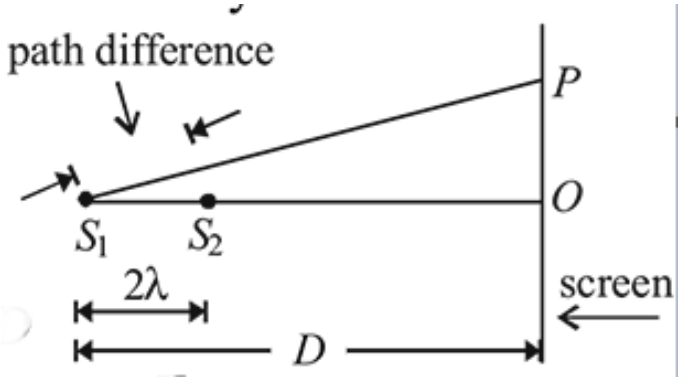
$D\sqrt{3}$

Solution:

Referring to the figure, the path difference between the two waves starting from S_1 **and** S_2 **turns out to be** $(2\lambda \cos\theta) = n\lambda$ **where** n **is taken as 1 to get the point of maximum intensity which is the same as a**

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$= \sqrt{3}$, giving $PO = D\sqrt{3}$.



Q.5 Answer:

$5500\mathring{A}$

Solution:

The path difference produced by a sheet $\Delta = (\mu - 1)t$

According to the given condition (for minimum thickness) $(\mu - 1)t = \frac{\lambda}{2}$

$\therefore \quad t = \frac{\lambda}{2(\mu-1)} = \frac{6600\times10^{-10}}{2(1.6-1)} = 5500\mathring{A}.$

Q.6 Answer:

6.738×10^{-5} cm

Solution:

In this case, both the rays suffere a phase change of 180° **and the conditions for destructive interference is** $2\mathrm{nd} = \left(\mathrm{m} + \frac{1}{2}\right)\lambda_1$ **and** $2\mathrm{nd} = \left(\mathrm{m} + \frac{3}{2}\right)\lambda_2$

$\frac{\mathrm{m}+\frac{1}{2}}{\mathrm{m}+\frac{3}{2}} = \frac{\lambda_2}{\lambda_1} = \frac{5000}{700} = \frac{5}{7}$

and $\mathrm{d} = \frac{\left(\mathrm{m}+\frac{1}{2}\right)\lambda_1}{2\mathrm{n}} = \frac{2.5\times7000}{2\times1.3}$

$= 6.738 \times 10^{-5}$ cm

Q.7 Answer:

100

Solution:

Constructive interference happens when $2t = (m - 1/2)\lambda$**. The minimum value for** $m =$ **is** $m = 1$**, the maximum value is the integer portion of** $\frac{2d}{\lambda} + \frac{1}{2} = \frac{2\times0.034\times10^{-3}}{680\times10^{-9}} + \frac{1}{2} = 100.5$

$m_{\max} = 100$

Q.8 Answer:

$\frac{2\pi}{\lambda}\left(\mu_1 \mathrm{L}_1 - \mu_2 \mathrm{L}_2\right)$

Solution:

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Correct option is (b) $\frac{2\pi}{\lambda}\left(\mu_1 \mathrm{L}_1-\mu_2 \mathrm{L}_2\right)$

Here one ray passes through the medium of refractive index μ_1 **and second ray passes through the medium of refractive index** μ_2**.**

Optical path of first ray $=\mu_1 \times \mathrm{L}_1$ **similarly**

Optical path of second ray $=\mu_2 \times \mathrm{L}_2$

Effective path difference is $\left(\mu_1 \mathrm{L}_1-\mu_2 \mathrm{L}_2\right)$

Effective phase difference is $\frac{2\pi}{\lambda}\left[\mu_1 \mathrm{L}_1-\mu_2 \mathrm{L}_2\right]$

Q.9 Answer:

1.6

Solution:

Change in path difference for any point on screen is $|\mu-1.8|\mathrm{t}$.

For central maxima, phase difference $=0$**. Hence,** $\mathrm{d} \sin \theta-|\mu-1.8| \mathrm{t}=0$

$\Rightarrow \mathrm{d}\theta=|\mu-1.8| \mathrm{t}$ [q **is very small and is in radian]**

$\Rightarrow \quad|\mu-1.8|=\frac{10^{-3} \times 0.1}{0.5 \times 10^{-3}}=0.2 \Rightarrow \mu=2$ **or** 1.6

Q.10 Answer:

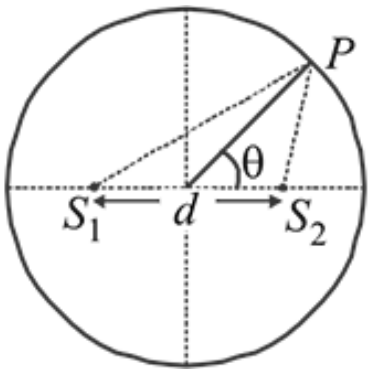
$\cos^{-1} \frac{4\lambda}{d}$

Solution:

Path difference. $\Delta x=S_1 P-S_2 P=d \cos \theta$

$\therefore d \cos \theta=4\lambda$

or $\cos \theta=(4\lambda/\mathrm{d})$



Q.11 Answer:

$4780 \mathring{A}$

Solution:

Fringe $\mathrm{Width}(\beta)=$ No of fringes **Space occupied**

$$\beta_1=\frac{20}{2.3} \quad \beta_2=\frac{30}{2.8}$$

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$$\beta = \frac{\lambda D}{d} \quad D \rightarrow \text{Space btw slits and screen}$$

$d \rightarrow$ Space blw slits

$$\frac{\beta_1}{\beta_2} = \frac{20}{2.3 \times 30} \times 2.8 = \frac{5890}{\lambda_2}$$

$$\lambda_2 = 4780 A$$

Q.12 Answer:

$49 : 1$

Solution:

Correct option is (a) $49 : 1$

From figure $\mathrm{I_1} = \frac{\mathrm{I}}{4}$ **and** $\mathrm{I_2} = \frac{9\mathrm{I}}{64} \Rightarrow \frac{\mathrm{I_2}}{\mathrm{I_1}} = \frac{9}{16}$

By using $\frac{\mathrm{I_{max}}}{\mathrm{I_{min}}} = \left(\frac{\sqrt{\frac{\mathrm{I_2}}{\mathrm{I_1}}}\mathrm{I_1}+1}{\sqrt{\frac{\mathrm{I_2}}{\mathrm{I_1}}}-1}\right)^2 = \left(\frac{\sqrt{\frac{9}{16}}+1}{\sqrt{\frac{9}{16}}-1}\right)^2 = \frac{49}{1}$

Q.13 Answer:

$4655 \mathring{A}$

Solution:

$$\beta = \frac{\lambda}{2\mu \tan \alpha} \simeq \frac{\lambda}{2\mu\alpha}$$

$$\therefore \lambda = \beta \times 2\mu\alpha$$

$$= 0.12 \times 10^{-2} \times 2 \times 1 \times \left(\frac{40}{60 \times 60} \times \frac{\pi}{180}\right)$$

$= 4655 \mathring{A}.$

Q.14 Answer:

20

Solution:

(a) Path difference at P **is** $\Delta \mathrm{x} = 2\left(\frac{\mathrm{x}}{2}\cos\theta\right) = \mathrm{x}\cos\theta$

ar intensity to be

$\Delta \mathrm{x} = \mathrm{n}\lambda$

$(\mathrm{n} = 0, 1, 2, 3, \ldots.)$

or $x\cos\theta = n\lambda$

or $\quad \cos\theta = \frac{\mathrm{n}\lambda}{\mathrm{x}} \geq 1$

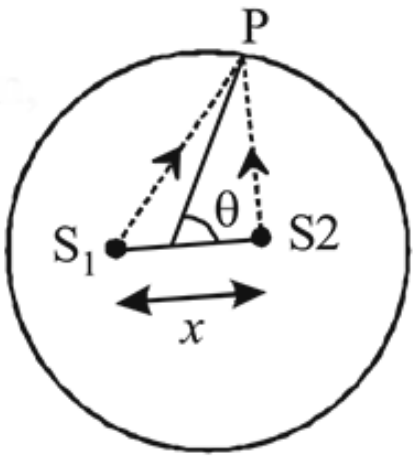
$\therefore \quad \mathrm{n} \geq \frac{\mathrm{x}}{\lambda}$

Subsituting $\mathrm{x} = 5\mathrm{l}$**, we get** $\mathrm{n} \geq 5$ **or** $\mathrm{n} = 1, 2, 3, 4, 5, \ldots\ldots$

Therefore in all four quadrants there can be 20 maxima. There are more maxima at $\theta = 90^\circ$

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and $\theta = 180^\circ$. But $\mathrm{n} = 5$ corresponds to $\theta = 90^\circ$ and $\theta = 270^\circ$ which are coming only twice while we have multuplies it four times. Therefore, total number of maxima are still 20 , i.e., $\mathrm{n} = 1$ to 4 in four quadrants (total 16) plus more at $\theta = 0^\circ, 90^\circ, 180^\circ$ and 270°.



Q.15 Answer:

10^{-7}

Solution:

Concept:-Approach to this kind of problems are like first find λ_{film} which equal to $\frac{\lambda}{\mu}$; here λ is the wavelength of light to be minimized reflection \& μ is refractive index of coating material.

$\lambda_{\text{film}} = \frac{\lambda}{\mu}$.

Then the thickness is equal to $\frac{\lambda_{\text{film}}}{\mu}$

i.e $t = \frac{\lambda_{\text{film}}}{\mu}$

Solution:- Given $\mu = 1.38 \& \lambda = 550$ nm

Hence $\lambda_{\text{film}} = \frac{\lambda}{\mu} = \frac{550}{1.38}$

$\lambda_{\text{film}} = 398.55$ nm

Now,thickness of coating , t is.

$\mathrm{t} = \frac{\lambda_{\text{film}}}{\mu} = \frac{398.55 \text{ nm}}{\mu}$

$\mathrm{t} \approx 100 \text{ nm} \approx 10^{-7} \text{ m}$

Q.16 Answer:

$\frac{4D\lambda}{d}$

Solution:

Due to reflection virtual source will be formed at distance D from mirror. The effective distance of the screen $= 2\mathrm{D} + 2D = 4D$

$\therefore$ Fringe width $= \frac{4D\lambda}{d}$

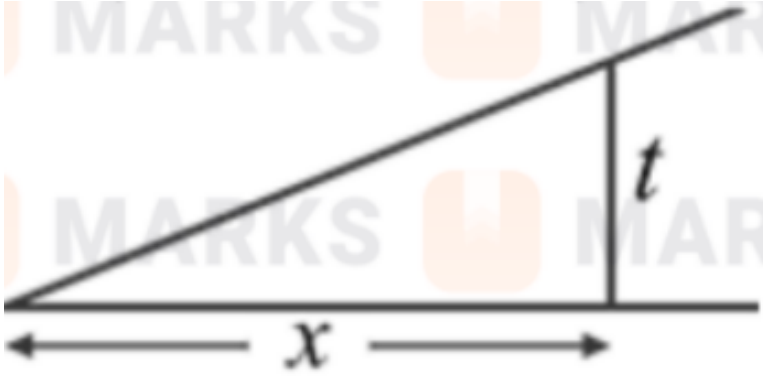
Q.17 Answer:

0.0055 cm

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Solution:

$$\beta = \frac{\lambda}{2\mu\alpha} = \frac{\lambda}{2\mu\left(\frac{t}{x}\right)}$$
$$= \frac{\lambda x}{2\mu t} \quad \text{or } t = \frac{\lambda x}{2\beta\mu}$$



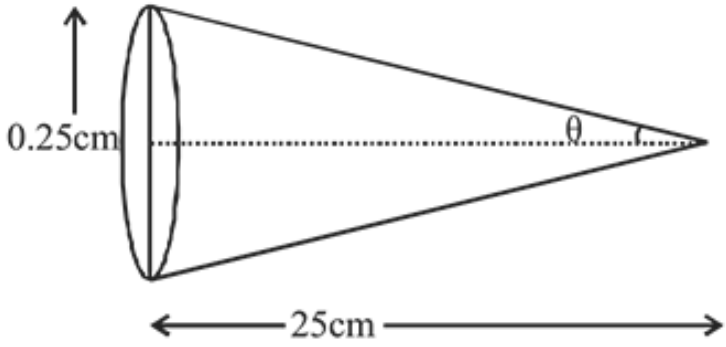
$$t = \frac{5500\times10^{-10}\times10\times10^{-2}}{2\times0.5\times10^{-3}\times1.5}$$
$$t = 0.0036 \times 10^{-2}\ \text{mKS}$$
$$t = 0.0036\ \text{cm}(2)$$

Q.18 Answer:

$30\mu\text{m}$

Solution:

$$\sin\theta = \frac{0.25}{25} = \frac{1}{100}$$



Resolving power $= \frac{1.22\lambda}{2\mu\sin\theta} = 30\mu\text{m}$

Q.19 Answer:

$1 : 49$

Solution:

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As shown in the figure, the interference wil. be between $0.25I = I_1$ and $0.14I = I_2$

$\frac{I_{max}}{I_{min}} = \frac{\left(\sqrt{I_1}+\sqrt{I_2}\right)^2}{\left(\sqrt{I_1}-\sqrt{I_2}\right)^2} = \frac{[\sqrt{0.25I}+\sqrt{0.14I}]^2}{[\sqrt{0.25I}-\sqrt{0.14I}]^2} = \frac{49}{1}$

Q.20 Answer:

Origin of spectra

Solution:

Huygen's construction of wavefront does not apply to origin of spectra which is explained by quantum theory.

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